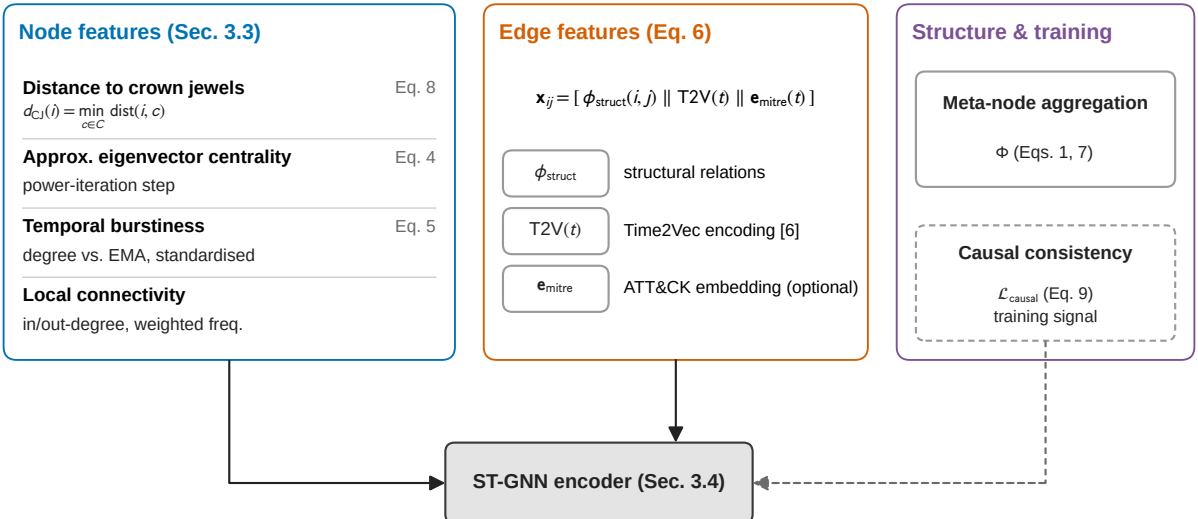
Node features (Sec. 3.3)
Distance to crown jewels
Eq. 8
$d_{CJ}(i) = \min_{c \in C} \mathrm{dist}(i, c)$
Approx. eigenvector centrality
Eq. 4
power-iteration step
Temporal burstiness
Eq. 5
degree vs. EMA, standardised
Local connectivity
in/out-degree, weighted freq.
Edge features (Eq. 6)
$\mathbf{x}_{ij} = [\, \phi_{\mathrm{struct}}(i, j) \,\|\, \mathrm{T2V}(t) \,\|\, \mathbf{e}_{\mathrm{mitre}}(t) \,]$
ϕ_{struct}
structural relations
T2V(t)
Time2Vec encoding [6]
$\mathbf{e}_{\mathrm{mitre}}$
ATT&CK embedding (optional)
Structure & training
Meta-node aggregation
Φ (Eqs. 1, 7)
Causal consistency
$\mathcal{L}_{\mathrm{causal}}$ (Eq. 9)
training signal
ST-GNN encoder (Sec. 3.4)